HR DATABASE MANAGEMENT SYSTEM

Project Title: HR Database Management System

Technology Used:

- SQL Server
- Azure Data Studio

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Introduction

Overview

The HR Database Management System is designed to manage and analyze organizational employee data. The database contains information related to employees, departments, jobs, locations, countries, regions, and dependents.

Objectives

- Store and manage employee information efficiently.
- Perform data analysis using SQL queries.
- Generate business insights from HR data.
- Understand relationships between different organizational entities.

Database Structure

Tables Used

1. Employees
2. Departments
3. Jobs
4. Locations
5. Countries
6. Regions
7. Dependents

Relationships

- Employees belong to Departments.
- Departments are located in Locations.
- Locations belong to Countries.
- Countries belong to Regions.
- Employees may have Dependents.
- Employees report to Managers.

Methodologies

SQL Concepts Applied

Data Retrieval

- SELECT
- DISTINCT
- TOP
- ORDER BY
- WHERE

Data Filtering

- LIKE
- BETWEEN
- IN
- EXISTS
- ANY
- ALL

Data Modification

- UPDATE
- ALTER TABLE

Data Integrity

- Primary Key
- Foreign Key
- Constraints

Approaches

Joins Used

INNER JOIN

Used to retrieve matching records from multiple tables.

LEFT JOIN

Used to retrieve all records from one table and matching records from another.

SELF JOIN

Used to identify manager-employee relationships.

FULL OUTER JOIN

Used to retrieve all matching and non-matching records.

CROSS JOIN

Used to generate all possible combinations between tables.

Key Findings

Employee Analysis

- Employees are distributed across multiple departments.
- Several departments have significantly larger headcounts.
- Salary distribution varies among departments.
- Some employees have dependents while others do not.
- Managerial hierarchy is clearly established through manager IDs.

Department Analysis

- Certain departments contribute higher salary expenses.
- Departments differ significantly in employee count.

Salary Analysis & Insights

Aggregation Analysis

Performed using:

- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()

Key Observations

- Highest paid employees were identified using subqueries.
- Department-wise average salaries were calculated.
- Total salary expenditure per department was analyzed.
- Departments with high salary costs were identified.

Business Value

Helps HR management:

- Monitor payroll expenses.
- Identify salary gaps.
- Support compensation planning.

Advanced SQL Analysis

Advanced Queries Performed

Subqueries

- Employees working in Location 1700.
- Employees earning above average salary.
- Highest salary employee identification.
- Department-wise salary comparisons.

EXISTS & NOT EXISTS

- Employees with dependents.
- Employees without dependents.

CASE Statements

Used to categorize employees based on salary levels:

- Low
- Average
- High

Actionable Recommendations

Workforce Management

- Review departments with low staffing levels.
- Monitor departments with excessive headcounts.

Compensation Planning

- Evaluate salary distribution across departments.
- Identify departments with unusually high payroll expenses.

Employee Engagement

- Track employees with long tenure.
- Recognize work anniversaries and achievements.

Data Governance

- Enforce foreign key constraints.
- Maintain data consistency across HR tables.

Conclusion

Project Summary

The HR Database Management System successfully demonstrated the practical application of SQL for data retrieval, filtering, aggregation, joins, and advanced analysis.

Skills Demonstrated

- SQL Query Writing
- Data Analysis
- Data Aggregation
- Reporting
- Business Insight Generation
- Relational Database Management

Outcome

The project provides meaningful HR insights that can support workforce planning, salary management, and organizational decision-making.

Thank You